INTRODUCTION TO ROBOTICS

Title: Homogeneous Transformation in Planar and Spatial Mechanisms using Python Robotics Toolbox
Name: K Prithvi
Registration No.: 24011101047

1. Aim

To implement homogeneous transformations for planar (SE2) and spatial (SE3) robotic mechanisms using Python Robotics Toolbox and SpatialMath, and to determine transformation matrices, final positions, orientations, and coordinate-frame visualizations.

2. Objectives

- Generate and display homogeneous transformation matrices.
- Perform planar translation and rotation using SE2.
- Perform spatial translation and rotations about X, Y and Z axes using SE3.
- Combine multiple transformations in a specified sequence.
- Study the effect of changing the order of transformations.
- Determine final position and orientation.
- Visualize transformed coordinate frames.
- Apply transformations to two-link planar and spatial robots.

3. Software and Libraries

Software / Library	Purpose
Python 3.10+	Programming environment
Robotics Toolbox for Python	Robotics transformations and visualization
SpatialMath Python	SE2 and SE3 rigid-body transformations
NumPy	Numerical calculations and matrix comparisons
Matplotlib	2D and 3D plotting
Jupyter Notebook / VS Code	Program development and execution

4. Theory

A homogeneous transformation matrix represents both the position and orientation of a rigid body in a single matrix. For a spatial transformation, the general form is:

$$\mathbf{T} = \begin{bmatrix} \mathbf{R} & \mathbf{P} \\ \mathbf{0} & \mathbf{1} \end{bmatrix}$$

where R is the 3×3 rotation matrix and P is the 3×1 position vector. The bottom row completes the homogeneous representation. For planar motion, SE2 represents a 3×3 homogeneous transformation; for spatial motion, SE3 represents a 4×4 transformation.

Transformation matrices can be multiplied to represent a sequence of operations. The order of multiplication is significant because rigid-body transformations are generally non-commutative. Therefore, changing the order can change both the final position and orientation.

5. Code

Source code link : https://github.com/KPR-24110081/Robotics/tree/main

6. Result

The homogeneous transformation experiment was successfully implemented for planar and spatial mechanisms. SE2 and SE3 transformations were used to perform translations, rotations, combined transformations, sequential transformations, and robot-link transformations. Final positions and orientations were obtained, coordinate frames were visualized, and the effect of changing transformation order was demonstrated.

7. Conclusion

This experiment provides practical understanding of homogeneous transformation matrices in robotics. The implementation shows how individual translation and rotation operations can be combined to determine the complete pose of a robotic link or end-effector. The two-link planar and spatial robot exercises further demonstrate how transformations can be chained to model robotic mechanisms. The comparison of different operation orders confirms that transformation multiplication is generally non-commutative and therefore the order of operations must be carefully maintained.